CURRICULUM VITA

Yilong Wang

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Education

- 2011 - 2018 Ph.D., Mathematics, The Ohio State University.
  Advisor: Thomas Kerler.
- 2011 - 2018 M.S., Mathematics, The Ohio State University.
- 2007-2011 B.S. (Hons.), Mathematics, Zhejiang University.

Employments

- 2026- Associate Professor, BIMSA.
- 2021-2026 Assistant Professor, BIMSA.
- 2018-2021 Postdoc Researcher, Louisiana State University.
  Mentor: Siu-Hung Ng.

Publications and preprints

1. Alterfold theory and topological modular invariance (with Zhengwei Liu, Shuang Ming and Jinsong Wu).

   Comm. Math. Phys. 407(5), no. 102, 2026.

2. Alterfold topological quantum field theory (with Zhengwei Liu, Shuang Ming and Jinsong Wu).

   Science China Mathematics, https://doi.org/10.1007/s11425-025-2574-6, 2026.

3. Non-semisimple open-closed 3d TFT (with Ingo Runkel).

   Preprint. arXiv:2608.08057.

4. Modular fusion categories with trivial Torelli group actions (with Liang Chang and Siu-Hung Ng).

   Preprint. arXiv:2608.04462.

5. On the gauge invariance of the Kuperberg invariant of certain high genus framed 3-manifolds (with Liang Chang and Saifei Zhai).

   Preprint. arXiv:2601.19485.

6. Generalized Witt and Morita equivalences (with Liang Kong and Hao Zheng).

   Preprint. arXiv:2511.02624.

7. Generalizations of Frobenius-Schur indicators from Kuperberg invariants (with Liang Chang and Siu-Hung Ng).

   Preprint. arXiv:2506.07409.

8. 3-Alterfolds and quantum invariants (with Zhengwei Liu, Shuang Ming and Jinsong Wu).

   Preprint. arXiv:2307.12284.

9. On symmetric representations of $\mathrm{SL}_2(\mathbb{Z})$ (with Siu-Hung Ng and Samuel Wilson).

   Proc. Amer. Math. Soc. 151, no. 4, 1415-1431, 2023.

10. The Witt classes of $\mathfrak{so}(2r)_{2r}$ (with Eric C. Rowell and Yuze Ruan).

    Comm. Algebra. 50 (2022), no. 12, 5246-5265.

11. Modular categories with transitive Galois actions (with Siu-Hung Ng and Qing Zhang).

    Comm. Math. Phys. 390, no. 3, 1271-1310, 2022.

12. Higher central charges and Witt groups (with Siu-Hung Ng, Eric C. Rowell and Qing Zhang).

    Adv. Math. 404, Paper No. 108388, 2022.

13. Classification of spherical fusion categories of Frobenius-Schur exponent 2 (with Zheyan Wan).

    Algebra Colloq. 28 (2021), no. 1, 39-50.

14. Higher Gauss sums of modular categories (with Siu-Hung Ng and Andrew Schopieray).

    Selecta Math. (N.S.) 25, no. 4, Paper No. 53, 32 pp, 2019.

15. On modular group representations associated to $SO(p)_2$-TQFTs.

    J. Knot Theory Ramifications 28, no. 5, 1950037, 20 pp, 2019.

16. On integrality of $\mathrm{SO}(n)$-Level-2 TQFTs.

    Thesis. The Ohio State University, 2018.

17. Random walk invariants from R-matrices (with Thomas Kerler).

    Algebr. Geom. Topol. 16 (2016), no. 1, 569-596.

Honors and Awards

1. Invited Speaker, International Congress of Chinese Mathematicians (ICCM), 2025.
2. Ruolin Award for Research, Tsinghua University, 2023.
3. Ruolin Award for Research, Tsinghua University, 2022.
4. Beijing Overseas Talent Program, 2022.
5. Special Graduate Assignments, The Ohio State University, 2015, 2016, 2017.

Grants

1. National Natural Science Foundation of China, Grant No. 12571041, 2026-2029.
2. National Natural Science Foundation of China, Grant No. 12301045, 2024-2026.

Conferences organized

- *Frontiers in Algebra*, BIMSA, April 2026.
- *International Workshop on Hopf Algebras and Tensor Categories*, Tsinghua Sanya International Mathematics Forum, January 2026.
- *Advances in Quantum Algebra*, Tsinghua Sanya International Mathematics Forum, February 2025.
- *Advances in Quantum Algebra*, BIMSA, January 2024.
- *Mini-workshop on knot theory and its applications*, BIMSA, July 2023.
- *AMS Special Session on Quantum Symmetries: Subfactors and Fusion Categories (a Mathematics Research Communities Special Session).* Joint Mathematics Meeting, January 2019, Baltimore, Maryland.

Selected Talks

1. *Generalizations of Frobenius-Schur indicators from Kuperberg invariants*, ICCM 2025, Shanghai, January 2026.
2. *From Frobenius-Schur indicators to Kuperberg invariants*, Quantum field theory with boundaries, impurities, and defects, Isaac Newton Institute for Mathematical Sciences, Cambridge, UK, September 2025.
3. *Algebraic properties of modular tensor categories and 3dTFTs*, Peaks, Depth and Waves: Mathematics in Asia, Xinjiang, August 2025.
4. *3-alterfolds and their TQFTs*, The 4th International Conference on Operad Theory and Related Topics, Nankai University, November 2024.

5. *Alterfold invariants and associated TQFTs*, The 9th Hopf Algebra Conference, Harbin Institute of Technology, August 2024.

6. *Gauge invariants from topology*, The 8th Hopf Algebra Conference, Guangzhou University, December 2023.

7. *Modular tensor categories from $SL_2(\mathbb{Z})$ representations*, Special Session on Hopf Algebras and Related Topics, Canadian Mathematics Society Summer Meeting, June 2023.

8. *Modular tensor categories from $SL_2(\mathbb{Z})$ representations*, Shanghai Jiao-Tong University, April 2022.

9. *Modular categories and their classifications*, Summer Frontier Research Lectures, Qingdao University, July 2022.

10. *Classification of transitive modular categories*, University Quantum Symmetries Lectures (UQSL), April 2021 (online).

11. *Modular categories with transitive Galois group actions*, FRG Seminar, September 2020 (online).

12. *Witt group invariants of modular categories*, Operator Algebra Seminar, University of California, Riverside, May 2020 (online).

13. *Algebraic properties of modular tensor categories*, Colloquium, Wayne State University, February 2020.

14. *Integrality of modular tensor categories*, Algebra seminar, University of Louisiana at Lafayette, November 2019.

15. *Classification of spherical fusion categories of Frobenius-Schur exponent 2*, Southern Regional Algebra conference, University of Louisiana at Lafayette, April 2019.

16. *On higher Gauss sums of modular categories*, Southern Regional Number Theory Conference, Louisiana State University, April 2019.

17. *Classification of spherical fusion categories of Frobenius-Schur exponent 2*, Quantum Algebra and Quantum Topology seminar, The Ohio State University, February 2019.

18. *Modular categories and RT-TQFTs*, Virtual Topology Seminar, Louisiana State University, September 2018.

19. *Higher Gauss sum and higher central charges of premodular fusion categories*, AMS Sectional Meeting Special Session on Quantum Symmetries, The Ohio State University, March 2018.

20. *Integrality for $SO(p)_2$-TQFTs for once-punctured torus*, Virtual Topology seminar, Louisiana State University, October 2017.

21. *Two constructions of the Jones polynomial*, Quantum Algebra and Quantum Topology seminar, The Ohio State University, September 2017.

22. *Integrality for $SO(p)_2$-TQFTs in genus 1*, AMS Sectional Meeting Special Session on Fusion Categories and Applications, Indiana University, Bloomington, April 2017.

23. *Metaplectic modular categories and the associated TQFT*, Quantum Algebra and Quantum Topology seminar, The Ohio State University, November 2016.

24. *Mapping class group representation from metaplectic modular categories and integrality*, Advances in Quantum and Low-dimensional topology, University of Iowa, March 2016.

25. *Random walk invariants of string links from R-matrices*, Knot Theory and Quantum Computation, UT Dallas, January 2015.

26. *Random walk invariants of string links via representation theory*, Knots in Washington, George Washington University, May 2014.

Teaching

BIMSA

- 2026 Nonsemisimple quantum invariants (Spring)
- 2025 Algebraic structures in nonsemisimple TFTs (Spring)
  From quadratic forms to modular tensor categories (Fall)
- 2024 Extensions of fusion categories (Spring)
  Extensions of fusion categories 2 (Fall)
- 2023 Topics on modular tensor categories (Spring)
  Pointed fusion categories and beyond (Fall)
- 2022 Introduction to algebraic number theory (Spring)
  Hopf algebras and tensor categories (Fall)
- 2021 Modular categories and Reshetikhin-Turaev TQFTs (Fall)

Louisiana State University

- Spring 2021 Math 2020 Solving Discrete Problems
- Fall 2020 Math 2070 Mathematical Methods in Engineering
- Fall 2019 Math 7290 Modular tensor categories and quantum invariants
- Fall 2019 Math 1550 Calculus I
- Fall 2018 Math 1550 Calculus I

The Ohio State University

- 2015-2017 Math 1152 Calculus II
- 2013-2015 Math 1172 Engineering Mathematics A
- 2012-2013 Math 1151 Calculus I

Other Activities

1. *Quantum field theory with boundaries, impurities, and defects*, Isaac Newton Institute for Mathematical Science, September 2025.

2. *TQFTs and their connections to representation theory and mathematical physics.* Hausdorff Center for Mathematics, 2023.

3. *Tensor categories and topological quantum field theories.* Mathematical Science Research Institute (MSRI), March 2020.

4. *Introductory workshop: Quantum symmetries.* Mathematical Science Research Institute (MSRI), January 2020.

5. *Quantum symmetries: Summer research program.* The Ohio State University, June 2019.

6. *Quantum symmetries: subfactors and fusion categories.* Mathematical Research Community (MRC) Program. University of Rhode Island, June 2018.

7. *Subfactors: planar algebras, quantum symmetries, and random matrices.* Mathematical Science Research Institute (MSRI), June 2017.